EXHAUST GAS TEMPERATURE SENSOR INSPECTION [SKYACTIV-D 1.5]

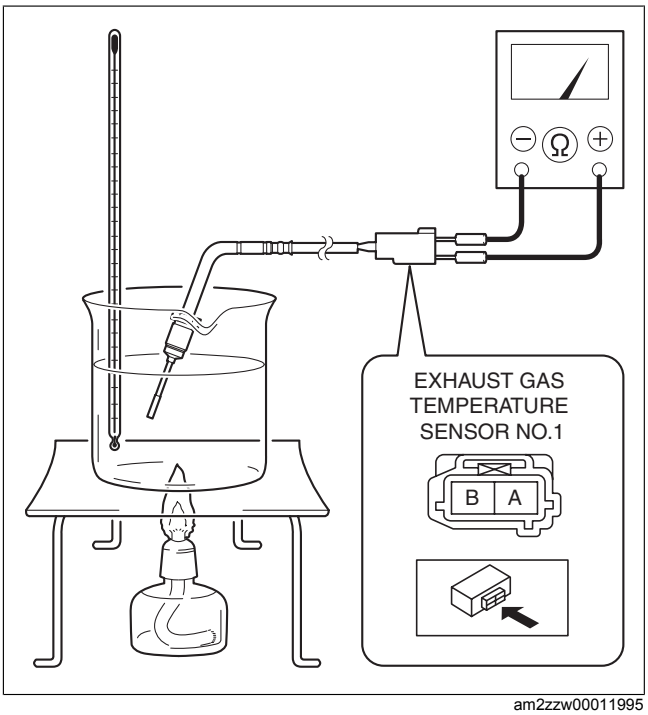
id0140q2446300

Exhaust Gas Temperature Sensor No.1

Warning
- **A hot engine and exhaust system can cause severe burns. Turn off the engine and wait until they are cool before removing the exhaust system.**

Caution
- **Perform exhaust gas temperature sensor removal/installation only when inspection or replacement is required, such as when DTCs for the sensor are displayed, because it may cause exhaust gas leaks.**
- **If removal/installation is performed, always verify that there is no exhaust gas leak after reinstallation.**

Resistance inspection
1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the following parts:
   (1) Engine cover (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   (2) Air cleaner (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   (3) Battery and battery tray (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   (4) Air inlet pipe, air hose and turbocharger air inlet hose (See INTAKE-AIR SYSTEM REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
   (5) Turbocharger solenoid valve (See TURBOCHARGER SOLENOID VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the exhaust gas temperature sensor No.1. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Place the exhaust gas temperature sensor No.1 in water with a thermometer, and heat the water gradually.



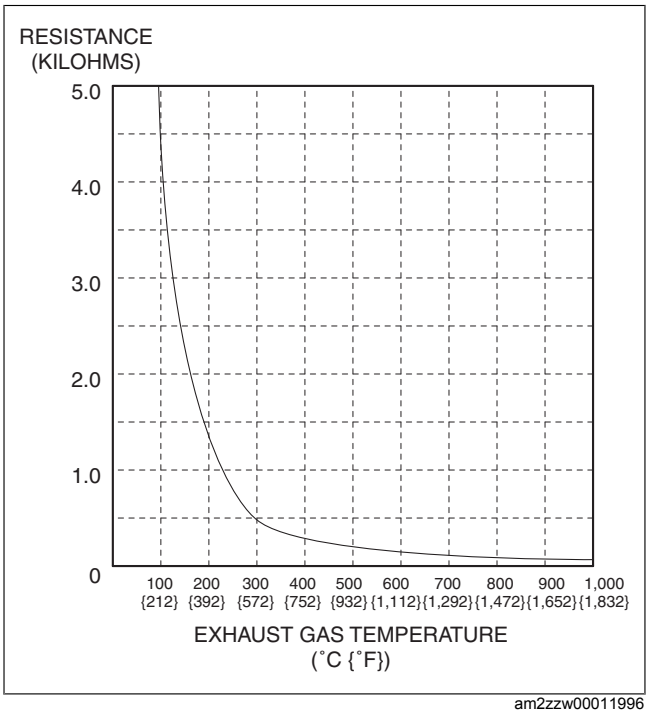
am2zzw00011995

5. Measure the resistance between exhaust gas temperature sensor No.1 terminals A and B.
   - If not as specified, replace the exhaust gas temperature sensor No.1. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

Resistance

Water temperature (°C {° F})	Resistance (Kilohms)
100 {212}	Approx. 4.51

Specification (Reference)



am2zzw00011996

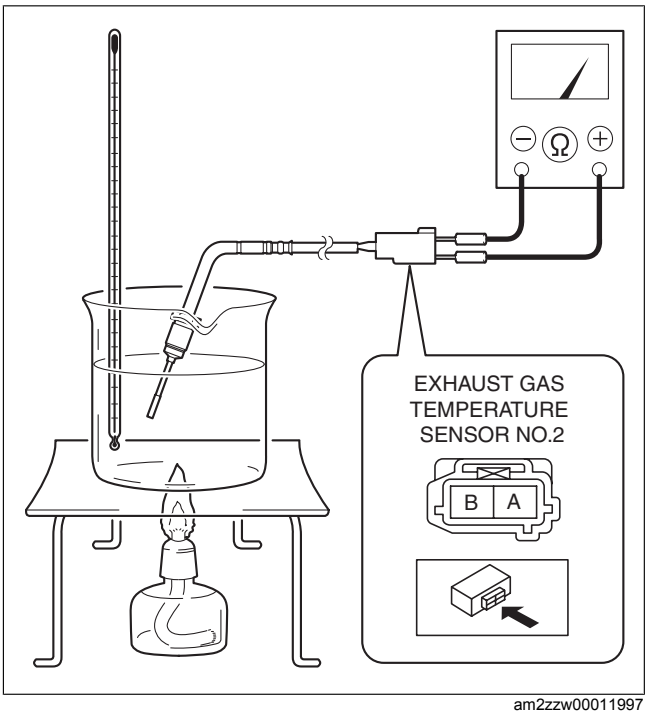
Exhaust Gas Temperature Sensor No.2

Warning
- **A hot engine and exhaust system can cause severe burns. Turn off the engine and wait until they are cool before removing the exhaust system.**

Caution
- **Perform exhaust gas temperature sensor removal/installation only when inspection or replacement is required, such as when DTCs for the sensor are displayed, because it may cause exhaust gas leaks.**
- **If removal/installation is performed, always verify that there is no exhaust gas leak after reinstallation.**

Resistance inspection
1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the exhaust gas temperature sensor No.2. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

4. Place the exhaust gas temperature sensor No.2 in water with a thermometer, and heat the water gradually.



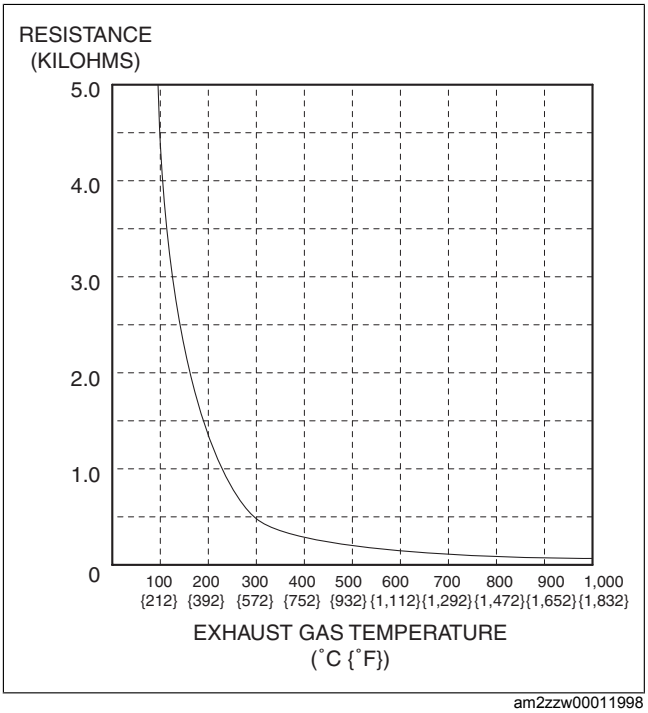
am2zzw00011997

5. Measure the resistance between exhaust gas temperature sensor No.2 terminals A and B.
   - If not as specified, replace the exhaust gas temperature sensor No.2. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

Resistance

Water temperature (°C {° F})	Resistance (Kilohms)
100 {212}	Approx. 4.51

Specification (Reference)



am2zzw00011998

Exhaust Gas Temperature Sensor No.3

Warning

- **A hot engine and exhaust system can cause severe burns. Turn off the engine and wait until they are cool before removing the exhaust system.**

Caution

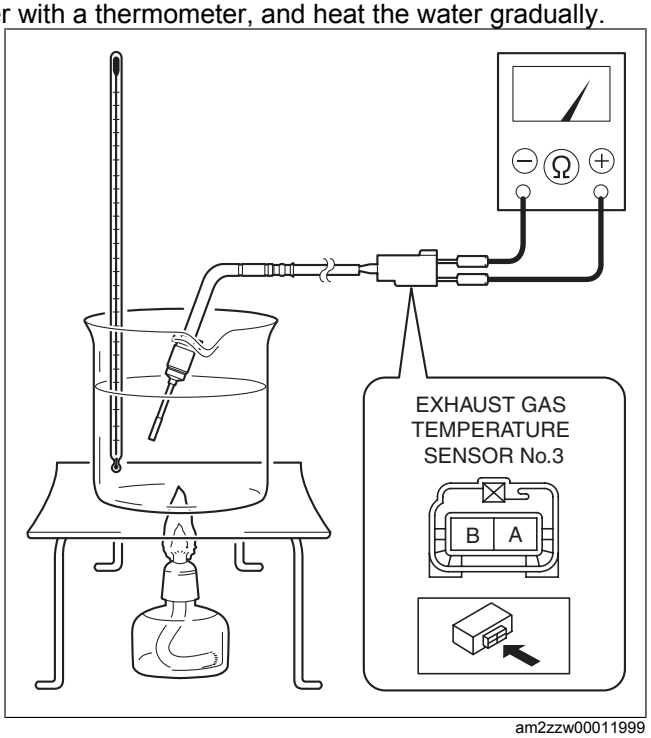
- **Do not allow flammable objects such as the taping for bundling wiring harness to come into contact with the catalytic converter.**
- **Perform exhaust gas temperature sensor removal/installation only when inspection or replacement is required, such as when DTCs for the sensor are displayed, because it may cause exhaust gas leaks.**
- **If removal/installation is performed, always verify that there is no exhaust gas leak after reinstallation.**

Resistance inspection

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Lift up the vehicle.
3. Remove the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
4. Remove the exhaust gas temperature sensor No.3. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5. Place the exhaust gas temperature sensor No.3 in water with a thermometer, and heat the water gradually.
6. Measure the resistance between exhaust gas temperature sensor No.3 terminals A and B.
   - If not as specified, replace the exhaust gas temperature sensor No.3. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)

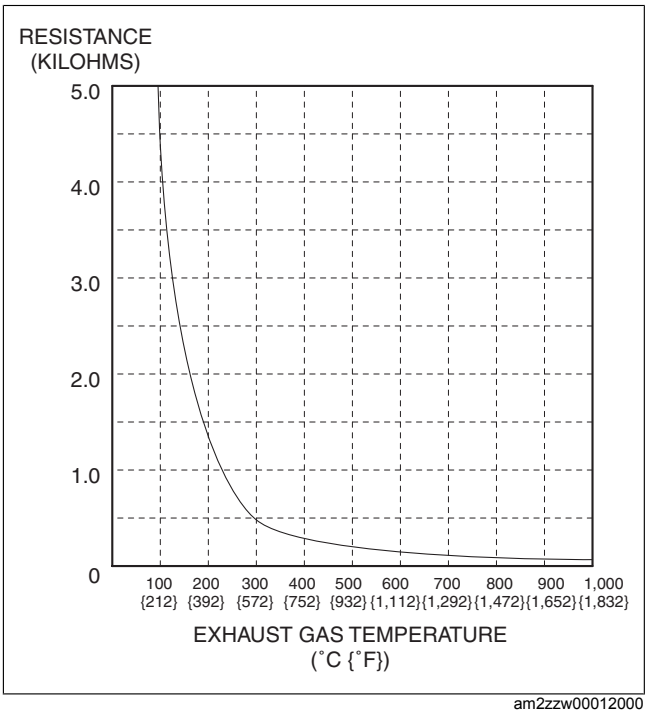
Resistance

Water temperature (°C {° F})	Resistance (Kilohms)
100 {212}	Approx. 4.51



am2zzw00011999

Specification (Reference)



am2zzw00012000

Exhaust Gas Temperature Sensor No.4

Warning
- **A hot engine and exhaust system can cause severe burns. Turn off the engine and wait until they are cool before removing the exhaust system.**

Caution
- **Do not allow flammable objects such as the taping for bundling wiring harness to come into contact with the catalytic converter.**
- **Perform exhaust gas temperature sensor removal/installation only when inspection or replacement is required, such as when DTCs for the sensor are displayed, because it may cause exhaust gas leaks.**
- **If removal/installation is performed, always verify that there is no exhaust gas leak after reinstallation.**

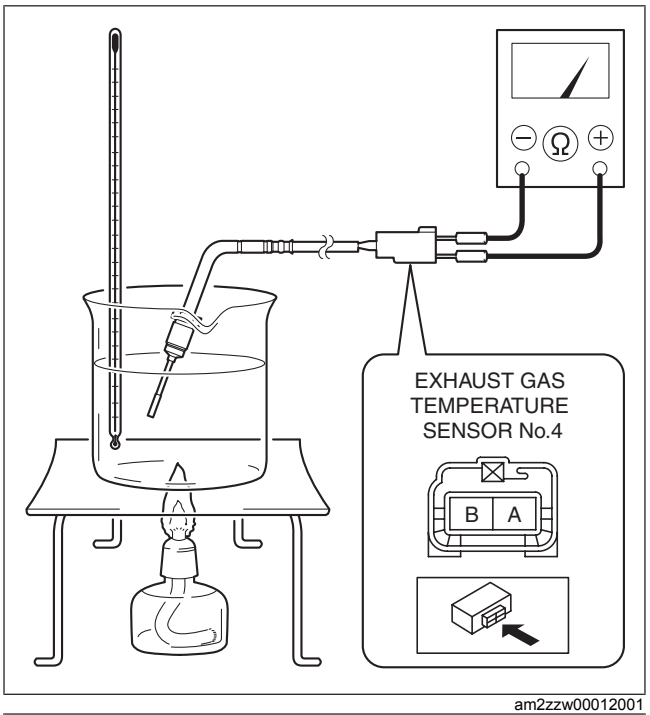
Resistance inspection
1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Lift up the vehicle.
3. Remove the plate (exhaust system). (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Remove the exhaust gas temperature sensor No.4. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5. Place the exhaust gas temperature sensor No.4 in water with a thermometer, and heat the water gradually.

6. Measure the resistance between exhaust gas temperature sensor No.4 terminals A and B.
   - If not as specified, replace the exhaust gas temperature sensor No.4. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)

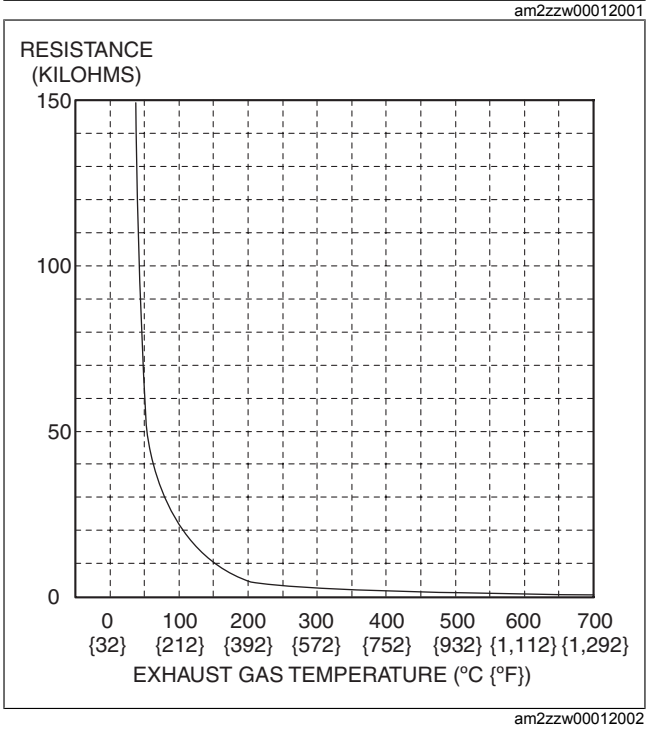
Resistance

Water temperature (°C {° F})	Resistance (Kilohms)
50 {122}	Approx. 53.27

Specification (Reference)



am2zzw00012001



am2zzw00012002